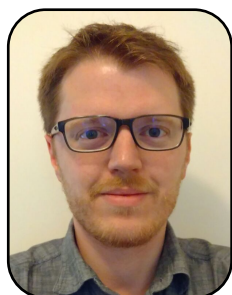


Guillaume Guex



June 3, 1982



Swiss nationality



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Summary:

Academic Researcher in applied mathematics, machine learning, computation linguistics, and digital humanities. Passionate about science, arts, computers and education. Appreciates data analysis, model creation and the implementation of numerical simulations. Easily works autonomously, constantly learns to use new tools and particularly likes to create synergies between various scientific fields.

« All models are wrong, but some are useful. »

George E. P. Box

WORKING EXPERIENCE

August 2024 – current

Senior lecturer (substitute) at the Department of Language and Information Sciences, University of Lausanne, Switzerland.

September 2019 – July 2024

Junior lecturer at the Department of Language and Information Sciences, University of Lausanne, Switzerland.

September 2021 – June 2023

Co-organizer of the 12th International Quantitative Linguistics Conference (QUALICO 2023), Lausanne, Switzerland.

September 2019 – January 2020

Substitute lecturer in quantitative textual analysis, Faculty of Social and Political Sciences, University of Lausanne, Switzerland.

March 2018 – August 2019

Methods specialist at Statistique Vaud, Département des Finances et des Relations Extérieures (DFIRE), Vaud, Switzerland.

September 2018 – January 2019

Lecturer in quantitative methods (20%) at the Faculty of Arts, University of Lausanne, Switzerland.

August 2016 – December 2017

Research assistant at the Institute of Information and Communication Technologies, Electronics and Applied Mathematics, Université Catholique de Louvain, Belgium.

March 2011 – February 2016

Teaching assistant at the Faculty of Arts and at the Faculty of Geosciences and Environment, University of Lausanne, Switzerland.

September 2014 – August 2015

Statistical expertise (mandate) for the Department of French as a Foreign Language, Faculty of Arts, University of Lausanne, Switzerland.

September 2011 – February 2012

Statistical expertise (mandate) for the Center of Gender Studies, University of Lausanne, Switzerland.

November 2008 – August 2009

Class teacher at the Junior High School of Belvédère, Lausanne, Switzerland.

August 2007 – June 2008

Substitute teacher (part time) at various Schools, Lausanne, Switzerland.

September 2000 – June 2015

Home tutoring (part time) in mathematics et physics, Lausanne, Switzerland.

EDUCATION

2011 – 2016 **Ph. D. in geography**, University of Lausanne.
2009 – 2011 **M. Sc. in statistics**, University of Neuchâtel.
2006 – 2008 28 ETS credits in applied mathematics, EPFL.
2001 – 2006 **B. Sc. In mathematics**, EPFL.
2000 – 2001 One semester in computer sciences, EPFL.
1997 – 2000 Swiss Matura Diploma, type C.

LANGUAGES

French C2 – Native proficiency
English C1 – Professional working proficiency
German B1 – Elementary proficiency

COMPUTER SKILLS

Operating systems	Linux, MacOS, Windows
General usage	Microsoft Office, Pages
Statistical programs	R, SPSS, SAS
Programming languages	Python, R, MATLAB, Julia, HTML, JavaScript, CSS, Java, C, C++
Others	Quarto, LaTeX, GIS, WordPress

PUBLICATIONS

Guex, G. (2025). "A framework for spatial clustering of textual objects: applications in topic clustering and text segmentation" *Cahiers du Centre de Linguistique et des Sciences du Langage*, (69), 73-96.

Jeanson, L., **Guex, G.**, & Xanthos, A. (2024). "Lexical diversity measurement using subsample entropy: formalism and evaluation". In *JADT 2024, mots comptés, textes déchiffrés, tome 1* (pp. 439-448). Anne Dister, Dominique Longrée.

Flaminio, S., Salomon Cavin, J., **Guex, G.**, & Moretti, M. (2024) "The urban in ecology: a quantitative textual analysis of the scientific literature over a century" *Urban Ecosystems*, 27(6), 2531-2542. DOI: <https://doi.org/10.1007/s11252-024-01603-4>

Pierre Leleux, P., Lebichot, B., **Guex, G.**, & Saerens M. (2023) "Sparse randomized policies for Markov decision processes based on Tsallis divergence regularization" *Knowledge-Based Systems*, vol. 300, p. 112105. DOI: <https://doi.org/10.1016/j.knosys.2024.112105>

Guex, G. (2023) "A Framework for Embedding Entities in a Textual Narrative: a Case Study on Les Misérables" in *Proceedings of COMHUM 2022*, CEUR-WS.

Guex, G., Loup, R., & Bavaud, F. (2023) "Estimation of flow trajectories in a multi-lines transportation network." in *Applied Network Science* 8, 44. DOI: <https://doi.org/10.1007/s41109-023-00570-7>

Courtain, S., **Guex, G.**, Kivimäki, I., & Saerens, M. (2023). "Relative entropy-regularized optimal transport on a graph: a new algorithm and an experimental comparison." in *International Journal of Machine Learning and Cybernetics*, 14(4), 1365-1390. DOI: <https://doi.org/10.1007/s13042-022-01704-6>

Guex, G., Courtain, S., & Saerens M. (2021) "Covariance and correlation measures on a graph in a generalized bag-of-paths formalism." in *Journal of Complex Networks*, Volume 8, Issue 6, DOI: <https://doi.org/10.1093/comnet/cnaa026>

Leleux, P., Courtain, S., **Guex, G.**, & Saerens, M. (2021) "Sparse randomized shortest paths routing with Tsallis divergence regularization." in *Data Mining and Knowledge Discovery*, 35, 986-1031. DOI: <https://doi.org/10.1007/s10618-021-00742-y>

Guex, G. & Xanthos, A. (2021) "Vers une estimation robuste des proportions lexicales" dans *Actes des 15èmes Journées Internationales d'Analyse statistique des Données Textuelles (JADT 2020)*. Peer-reviewed.

Courtain, S., Leleux, P., Kivimäki, I., **Guex, G.**, & Saerens, M. (2021). "Randomized shortest paths with net flows and capacity constraints." *Information Sciences*, 556, 341-360. DOI: <https://doi.org/10.1016/j.ins.2020.10.005>

Guex, G. & Guex, S. (2019) "La dette publique n'a pas d'impact sur la croissance et les taux d'intérêt" in *La Vie économique*. SECO.

Guex, G., Kivimäki, I., & Saerens, M. (2019). "Randomized optimal transport on a graph: framework and new distance measures." *Network Science*, 7(1), 88-122. DOI: <https://doi.org/10.1017/nws.2018.29>

Guex, G. & Guex, S. (2018) "Debt, economic growth and interest rates: An empirical study of the Swiss case, presenting a new long-term dataset: 1894-2014" in *Swiss Journal of Economics and Statistics*. DOI: [10.1186/s41937-017-0007-6](https://doi.org/10.1186/s41937-017-0007-6)

Emmanouilidis T., **Guex, G.**, & Bavaud F. (2016) "The transformed optimal transportation problem: sensitivity and segregation of the children-to-school constrained assignment in Lausanne" in *Proceedings of Spatial Accuracy 2016* (pp.264-277).

Guex, G. (2016) "Interpolating between Random Walks and Optimal Transportation Routes: flow with multiple sources and targets" in *Physica A* (vol. 450, pp.264-277), Elsevier. DOI: [10.1016/j.physa.2015.12.117](https://doi.org/10.1016/j.physa.2015.12.117)

Xanthos, A. & **Guex, G.** (2015) "On the robust measurement of inflectional diversity" in *Recent Contributions to Quantitative Linguistics* (vol. 70, pp. 241-254), Walter de Gruyter GmbH & Co KG.

Guex, G. & Bavaud, F. (2015) "Flow-Based Dissimilarities: Shortest Path, Commute Time, Max-Flow and Free Energy" in *Data Science, Learning by Latent Structures, and Knowledge Discovery* (pp. 101-111), Springer Berlin Heidelberg. Part of ISBN: [978-3-662-44983-7](https://doi.org/10.1007/978-3-662-44983-7)

Guex, G. (2014) "Spatial Graphs Cost and Efficiency: Exploring Edges Competition by MCMC" in *Geographic Information Science 2014* (pp. 97-108), Springer International Publishing. DOI: [10.1007/978-3-319-11593-1_7](https://doi.org/10.1007/978-3-319-11593-1_7)

Bavaud, F. & **Guex G.** (2012) "Interpolating between random walks and shortest paths: a path functional approach" in *Social Informatics* (pp. 68-81). Springer Berlin Heidelberg. Part of ISBN: [978-3-642-35386-4](https://doi.org/10.1007/978-3-642-35386-4)

THESES AND REPORTS

Guex, G. (2016) "Réseaux Spatiaux : modèles, classification et flux". Doctoral thesis, University of Lausanne. Supervisor: Bavaud, F.

Guex, G. (2011) "L'utilisation des transformations de Schoenberg dans le cadre de l'analyse factorielle". Master thesis, University of Neuchâtel. Supervisor: Matei, A.

Guex, G. (2007) "L'utilisation des chemins auto-évitant dans le cas de la percolation par sites : Théorie et Simulation Numériques". Scientific report, EPFL. Supervisor: Mountford, T.

Guex, G. (2007) "Psychophysics of perceptual learning". Scientific report, EPFL. Supervisor: Mast, F.

Guex, G. (2006) "Le mouvement Brownien Unidimensionnel : Théorie et Simulation Numériques". Scientific report, EPFL. Supervisor: Dalang, R.